

Mark Scheme

Q1.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$y = a + kx$, where a and k are constants	B1	This mark is given for stating a correct general equation
(b)	$500 = 800 \times 2 - (a + 800k)$ $-80 = 300 \times 2 - (a + 300k)$	M1	This mark is given for modelling the profit on the two days when bars of soap are sold for £2
	$a + 800k = 1100$ $a + 300k = 680$	M1	This mark is given for forming a pair of simultaneous equations to find values for a and k
	$500k = 420 \Rightarrow k = \frac{420}{500} = 0.84$ $a + (800 \times 0.84) = 1100$ $a = 1100 - 672 = 428$ Thus $y = 0.84x + 428$	A1	This mark is given for finding the values of a and k to show $y = 0.84x + 428$ as required
(c)	0.84 represents the cost of making one extra bar of soap in £s (i.e. 84p)	B1	This mark is given for a valid interpretation of the significance of 0.84
(d)	For n bars of soap $2n - (428 + 0.84n) > 0$	M1	This mark is given for a method to find the number of bars of soap to be made
	$1.16n - 428 > 0$ $n - \frac{428}{1.16} > 0$ $n = 369$ bars of soap	A1	This mark is given for correctly finding the number of bars of soap to be made
(Total 7 marks)			

Q2.

Question	Scheme	Marks	AOs
(a)	C is $(x-r)^2 + (y-r)^2 = r^2$ or $x^2 + y^2 - 2rx - 2ry + r^2 = 0$	B1	2.2a
	$y = 12 - 2x$, $x^2 + y^2 - 2rx - 2ry + r^2 = 0$ $\Rightarrow x^2 + (12 - 2x)^2 - 2rx - 2r(12 - 2x) + r^2 = 0$ or	M1	1.1b

	$y = 12 - 2x, \quad (x - r)^2 + (y - r)^2 = r^2$ $\Rightarrow (x - r)^2 + (12 - 2x - r)^2 = r^2$		
	$x^2 + 144 - 48x + 4x^2 - 2rx - 24r + 4rx + r^2 = 0$ $\Rightarrow 5x^2 + (2r - 48)x + (r^2 - 24r + 144) = 0 \quad *$	A1*	2.1
		(3)	
(b)	$b^2 - 4ac = 0 \Rightarrow (2r - 48)^2 - 4 \times 5 \times (r^2 - 24r + 144) = 0$	M1	3.1a
	$r^2 - 18r + 36 = 0 \text{ or any multiple of this equation}$	A1	1.1b
	$\Rightarrow (r - 9)^2 - 81 + 36 = 0 \Rightarrow r = \dots$	dM1	1.1b
	$r = 9 \pm 3\sqrt{5}$	A1	1.1b
		(4)	
(7 marks)			

Notes:

- (a)
- B1:** Deduces the correct equation of the circle
- M1:** Attempts to form an equation with terms of the form x^2 , x , r^2 , and xr only using $y = 12 \pm 2x$ and their circle equation which must be of an appropriate form. I.e. includes or implies an x^2, y^2, r^2 such as $x^2 + y^2 = r^2$
- If their circle equation starts off as e.g. $(x \pm a)^2 + (y \pm b)^2 = r^2$ then the B mark and the M mark can be awarded when the "a" and "b" are replaced by r or $-r$ as appropriate for their circle equation.
- A1*:** Uses correct and accurate algebra leading to the given solution.
- (b)
- M1:** Attempts to use $b^2 - 4ac \dots 0$ o.e. with $a = 5, b = 2r - 48, c = r^2 - 24r + 144$ and where ... is "=" or any inequality
- Allow minor slips when copying the a, b and c provided it does not make the work easier and allow their a, b and c if they are similar expressions.
- FYI $(2r - 48)^2 - 4 \times 5 \times (r^2 - 24r + 144) = 4r^2 - 192r + 2304 - 20r^2 + 480r - 2880 = -16r^2 + 288r - 576$
- A1:** Correct quadratic equation in r (or inequality). Terms need not be all one side but must be collected.
- E.g. allow $r^2 - 18r = -36$ and allow any multiple of this equation (or inequality).
- dM1:** Correct attempt to solve their 3TQ in r . Dependent upon previous M
- A1:** Careful and accurate work leading to both answers in the required form (must be simplified surds)

Q3.

Question	Scheme	Marks	AOs
(a)	$S_n = a + ar + ar^2 + \dots + ar^{n-1}$	B1	1.2
	$rS_n = ar + ar^2 + ar^3 + \dots + ar^n \Rightarrow S_n - rS_n = \dots$	M1	2.1
	$S_n - rS_n = a - ar^n$	A1	1.1b

	$S_n(1-r) = a(1-r^n) \Rightarrow S_n = \frac{a(1-r^n)}{(1-r)}$	A1*	2.1
		(4)	
(b)	$\frac{a(1-r^{10})}{1-r} = 4 \times \frac{a(1-r^5)}{1-r}$ or $4 \times \frac{a(1-r^{10})}{1-r} = \frac{a(1-r^5)}{1-r}$ Equation in r^{10} and r^5 (and possibly $1-r$)	M1	3.1a
	$1-r^{10} = 4(1-r^5)$	A1	1.1b
	$r^{10} - 4r^5 + 3 = 0 \Rightarrow (r^5 - 1)(r^5 - 3) = 0 \Rightarrow r^5 = \dots$ or e.g. $1-r^{10} = 4(1-r^5) \Rightarrow (1-r^5)(1+r^5) = 4(1-r^5) \Rightarrow r^5 = \dots$	dM1	2.1
	$r = \sqrt[5]{3}$ oe only	A1	1.1b
		(4)	
(8 marks)			

Notes:

- (a)
- B1:** Writes out the sum or lists terms. Condone the omission of S .
The sum must include the first and last terms and (at least) two other correct terms and no incorrect terms e.g. ar^n
Note that the sum may be seen embedded within their working.
- M1:** For the key step in attempting to multiply the first series by r and subtracting.
- A1:** $S_n - rS_n = a - ar^n$ either way around but condone one side being prematurely factorised (but not both) following correct work but this could follow B0 if insufficient terms were shown.
- A1*:** Depends on all previous marks. Proceeds to given result showing all steps including seeing both sides unfactorised at some point in their working.
Note: If terms are listed rather than added then allow the first 3 marks if otherwise correct but withhold the final mark.
- (b)
- M1:** For the correct strategy of producing an equation in just r^{10} and r^5 (and possibly $(1-r)$) with the “4” on either side using the result from part (a) and makes progress to at least cancel through by a
Some candidates retain the “ $1-r$ ” and start multiplying out e.g. $(1-r)(1-r^{10})$ and this mark is still available as long as there is progress in cancelling the “ a ”.
- A1:** Correct equation with the a and the $1-r$ cancelled. Allow any correct equation in just r^5 and r^{10}
- dM1:** Depends on the first M. Solves as far as $r^5 = \dots$ by solving a 3 term quadratic in r^5 by a valid method – see general guidance or by difference of 2 squares – see above
- A1:** $r = \sqrt[5]{3}$ oe only. The solution $r = 1$ if found must be rejected here.

(b) **Note:** For candidates who use $S_5 = 4S_{10}$ expect to see:

$$4 \times \frac{a(1-r^{10})}{1-r} = \frac{a(1-r^5)}{1-r} \Rightarrow 4(1-r^{10}) = (1-r^5) \text{ M1A0}$$

$$4r^{10} - r^5 - 3 = 0 \Rightarrow (4r^5 + 3)(r^5 - 1) = 0 \Rightarrow r^5 = \dots \text{ or } 4(1-r^5)(1+r^5) = (1-r^5) \Rightarrow r^5 = \dots \text{ dM1A0}$$

Example for (a)

$$S_n = a + ar + ar^2 + ar^3 + \dots + ar^{n-2} + ar^{n-1}$$

$$rS_n = ar + ar^2 + ar^3 + ar^4 + \dots + ar^{n-1} + ar^n$$

$$S_n - rS_n = a(1-r^n)$$

$$S_n(1-r) = a(1-r^n)$$

This scores B1M1A1A0:

B1: Writes down the sum including first and last terms and at least 2 other correct terms and no incorrect terms

M1: Multiplies by r and subtracts

A1: Correct equation (we allow one side to be prematurely factorised)

A0: One side was prematurely factorised

Q4.

Question	Scheme	Marks	AOs
	Attempts equation of line Eg Substitutes $(-2, 13)$ into $y = mx + 25$ and finds m	M1	1.1b
	Equation of l is $y = 6x + 25$	A1	1.1b
	Attempts equation of C Eg Attempts to use the intercept $(0, 25)$ within the equation $y = a(x \pm 2)^2 + 13$, in order to find a	M1	3.1a
	Equation of C is $y = 3(x+2)^2 + 13$ or $y = 3x^2 + 12x + 25$	A1	1.1b
	Region R is defined by $3(x+2)^2 + 13 < y < 6x + 25$ o.e.	B1ft	2.5
		(5)	
(5 marks)			

The first two marks are awarded for finding the equation of the line

M1: Uses the information in an attempt to find an equation for the line l .

E.g. Attempt using two points: Finds $m = \pm \frac{25-13}{2}$ and uses one of the points in their $y = mx + c$ or equivalent to find c . Alternatively uses the intercept as shown in main scheme.

A1: $y = 6x + 25$ seen or implied. This alone scores the first two marks. Do not accept $l = 6x + 25$

It must be in the form $y = \dots$ but the correct equation can be implied from an inequality. E.g. $\dots < y < 6x + 25$

The next two marks are awarded for finding the equation of the curve

M1: A complete method to find the constant a in $y = a(x \pm 2)^2 + 13$ or the constants a, b in $y = ax^2 + bx + 25$.

An alternative to the main scheme is deducing equation is of the form $y = ax^2 + bx + 25$ and setting and solving a pair of simultaneous equations in a and b using the point $(-2, 13)$ the gradient being 0 at $x = -2$. Condone slips. Implied by $C = 3x^2 + 12x + 25$ or $3x^2 + 12x + 25$

FYI the correct equations are $13 = 4a - 2b + 25$ ($2a - b = -6$) and $-4a + b = 0$

A1: $y = 3(x+2)^2 + 13$ or equivalent such as $y = 3x^2 + 12x + 25$, $f(x) = 3(x+2)^2 + 13$.

Do not accept $C = 3x^2 + 12x + 25$ or just $3x^2 + 12x + 25$ for the A1 but may be implied from an inequality or from an attempt at the area, E.g. $\int 3x^2 + 12x + 25 \, dx$

B1ft: Fully defines the region R . Follow through on their equations for l and C .

Allow strict or non-strict inequalities as long as they are used consistently.

E.g. Allow for example " $3(x+2)^2 + 13 < y < 6x + 25$ " " $3(x+2)^2 + 13 \leq y \leq 6x + 25$ "

Allow the inequalities to be given separately, e.g. $y < 6x + 25, y > 3(x+2)^2 + 13$. Set notation may be used so

$\{(x, y) : y > 3(x+2)^2 + 13\} \cap \{(x, y) : y < 6x + 25\}$ is fine but condone with or without any of $(x, y) \leftrightarrow y \leftrightarrow x$

Incorrect examples include " $y < 6x + 25$ or $y > 3(x+2)^2 + 13$ ", $\{(x, y) : y > 3(x+2)^2 + 13\} \cup \{(x, y) : y < 6x + 25\}$

Values of x could be included but they must be correct. So $3(x+2)^2 + 13 < y < 6x + 25, x < 0$ is fine

If there are multiple solutions mark the final one.

Q5.

Question	Scheme	Marks	AOs
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(a)	$2\log(4-x) = \log(4-x)^2$	B1	1.2
	$2\log(4-x) = \log(x+8) \Rightarrow \log(4-x)^2 = \log(x+8)$ $(4-x)^2 = (x+8)$ or $2\log(4-x) = \log(x+8) \Rightarrow \log(4-x)^2 - \log(x+8) = 0$ $\frac{(4-x)^2}{(x+8)} = 1$	M1	1.1b
	$16 - 8x + x^2 = x + 8 \Rightarrow x^2 - 9x + 8 = 0$ *	A1*	2.1
		(3)	
(a) Alternative - working backwards:			
	$x^2 - 9x + 8 = 0 \Rightarrow (4-x)^2 - x - 8 = 0$	B1	1.2
	$\Rightarrow (4-x)^2 = x + 8$ $\Rightarrow \log(4-x)^2 = \log(x+8)$	M1	1.1b
	$\Rightarrow 2\log(4-x) = \log(x+8)$ * Hence proved.	A1	2.1
(b)	(i) $(x =) 1, 8$	B1	1.1b
	(ii) 8 is not a solution as $\log(4-8)$ cannot be found	B1	2.3
		(2)	
(5 marks)			

Notes:

(a)

B1: States or uses $2\log(4-x) = \log(4-x)^2$

M1: Correct attempt at eliminating the logs to form a quadratic equation in x .

Note that this may be implied by e.g. $\log \frac{(4-x)^2}{(x+8)} = 0 \Rightarrow (4-x)^2 = x+8$

A1*: Proceeds to the given answer with at least one line where the $(4-x)^2$ has been multiplied out.

There must be no errors or omissions but condone invisible brackets around the arguments of the logs e.g. allow $\log 16 - 8x + x^2$ for $\log(16 - 8x + x^2)$ and $\log x + 8$ for $\log(x+8)$

Note we will allow a start of $(4-x)^2 = x+8$ with no previous work for full marks.

Some examples of how to mark (a) in particular cases:

$$2\log(4-x) = \log(x+8) \Rightarrow \log(4-x)^2 = \log(x+8) \Rightarrow \frac{\log(4-x)^2}{\log(x+8)} = 1$$

$$2\log(4-x) = \log(x+8) \Rightarrow \log(4-x)^2 - \log(x+8) = 0 \Rightarrow (4-x)^2 - x - 8 = 0$$

$$\Rightarrow 16 - 8x + x^2 - x - 8 \Rightarrow x^2 - 9x + 8 = 0$$

Scores **B1M1A1**

$$2\log(4-x) = \log(x+8) \Rightarrow \log(4-x)^2 - \log(x+8) = 0 \Rightarrow \frac{\log(4-x)^2}{\log(x+8)} = 0$$

$$\Rightarrow \frac{(4-x)^2}{(x+8)} = 1 \Rightarrow 16 - 8x + x^2 = x + 8 \Rightarrow x^2 - 9x + 8 = 0$$

Scores **B1M0A0**

(a) Alternative:

B1: Writes $x^2 - 9x + 8 = 0$ as $(4 - x)^2 - x - 8 = 0$ or equivalent

M1: Proceeds correctly to reach $\log(4 - x)^2 = \log(x + 8)$

A1: Obtains $2\log(4 - x) = \log(x + 8)$ and makes a (minimal) conclusion e.g. hence proved, QED, #, square etc.

(b)

B1: Writes down $(x =) 1, 8$

B1: Chooses 8 (no follow through here) and gives a reason why it should be rejected by referring to logs and which log it is.

They must refer to the 8 as the required value but allow e.g. $x \neq 8$ and there must be a reference to $\log(4 - x)$ or log of lhs or $\log(-4)$ or the $4 - 8$. Some acceptable reasons are: $\log(-4)$ can't be found/worked out/is undefined, $\log(-4)$ gives math error, $\log(-4) = \text{n/a}$, lhs is log(negative) so reject, you can't do the log of a negative number which would happen with $4 - 8$

Do not allow "you can't have a negative log" unless this is clarified further and do not allow "you get a math error" in isolation

There must be no contradictory statements.

Note that this is an independent mark but must have $x = 8$ (i.e. may have solved to get $x = -1, 8$ for first B mark)

Q6.

Question	Scheme	Marks	AOs
(a)	Either $3y^2 \rightarrow 4y \frac{dy}{dx}$ or $2xy \rightarrow 2x \frac{dy}{dx} + 2y$	M1	2.1
	$2x - 2x \frac{dy}{dx} - 2y + 6y \frac{dy}{dx} = 0$	A1	1.1b
	$(6y - 2x) \frac{dy}{dx} = 2y - 2x$	M1	2.1
	$\frac{dy}{dx} = \frac{2y - 2x}{6y - 2x} = \frac{y - x}{3y - x} *$	A1*	1.1b
		(4)	
(b)	$\left(\text{At } P \text{ and } Q \frac{dy}{dx} \rightarrow \infty \Rightarrow \right)$ Deduces that $3y - x = 0$	M1	2.2a
	Solves $y = \frac{1}{3}x$ and $x^2 - 2xy + 3y^2 = 50$ simultaneously	M1	3.1a
	$\Rightarrow x = (\pm) 5\sqrt{3}$ OR $\Rightarrow y = (\pm) \frac{5}{3}\sqrt{3}$	A1	1.1b
	Using $y = \frac{1}{3}x \Rightarrow x = ..$ AND $y = ..$	dM1	1.1b
	$P = \left(-5\sqrt{3}, -\frac{5}{3}\sqrt{3} \right)$	A1	2.2a
		(5)	
(c)	Explains that you need to solve $y = x$ and $x^2 - 2xy + 3y^2 = 50$ simultaneously and choose the positive solution	B1ft	2.4
		(1)	
(10 marks)			

Notes:

(a)

M1: For selecting the appropriate method of differentiating either $3y^2 \rightarrow 4y \frac{dy}{dx}$ or $2xy \rightarrow 2x \frac{dy}{dx} + 2y$

It may be quite difficult awarding it for the product rule but condone $-2xy \rightarrow -2x \frac{dy}{dx} + 2y$ unless you see evidence that they have used the incorrect law $vu' - uv'$

A1: Fully correct derivative $2x - 2x \frac{dy}{dx} - 2y + 6y \frac{dy}{dx} = 0$

Allow attempts where candidates write $2xdx - 2xdy - 2ydx + 6ydy = 0$

but watch for students who write $\frac{dy}{dx} = 2x - 2x \frac{dy}{dx} - 2y + 6y \frac{dy}{dx}$ This, on its own, is A0 unless you are

convinced that this is just their notation. Eg $\frac{dy}{dx} = 2x - 2x \frac{dy}{dx} - 2y + 6y \frac{dy}{dx} = 0$

M1: For a valid attempt at making $\frac{dy}{dx}$ the subject, with two terms in $\frac{dy}{dx}$ coming from $3y^2$ and $2xy$

Look for $(\dots \pm \dots) \frac{dy}{dx} = \dots$ It is implied by $\frac{dy}{dx} = \frac{2y-2x}{6y-2x}$

This cannot be scored from attempts such as $\frac{dy}{dx} = 2x - 2x \frac{dy}{dx} - 2y + 6y$ which only has one correct term.

A1*: $\frac{dy}{dx} = \frac{y-x}{3y-x}$ with no errors or omissions.

The previous line $\frac{dy}{dx} = \frac{2y-2x}{6y-2x}$ or equivalent must be seen.

(b)

M1: Deduces that $3y - x = 0$ or

M1: Attempts to find either the x or y coordinates of P and Q by solving their $y = \frac{1}{3}x$ with

$x^2 - 2xy + 3y^2 = 50$ simultaneously. Allow for finding a quadratic equation in x or y and solving to find at least one value for x or y .

This may be awarded when candidates make the numerator = 0 ie using $y = x$

A1: $\Rightarrow x = (\pm)5\sqrt{3}$ OR $\Rightarrow y = (\pm)\frac{5}{3}\sqrt{3}$

dM1: Dependent upon the previous M, it is for finding the y coordinate from their x (or vice versa)

This may also be scored following the numerator being set to 0 ie using $y = x$

A1: Deduces that $P = \left(-5\sqrt{3}, -\frac{5}{3}\sqrt{3}\right)$ OE. Allow to be $x = \dots$ $y = \dots$

(c)

B1ft: Explains that this is where $\frac{dy}{dx} = 0$ and so you need to solve $y = x$ and $x^2 - 2xy + 3y^2 = 50$

simultaneously and choose the positive solution (or larger solution).

Allow a follow through for candidates who mix up parts (b) and (c)

Alternatively candidates could complete the square $(x-y)^2 + 2y^2 = 50$ and state that y would reach a maximum value when $x = y$ and choose the positive solution from $2y^2 = 50$

Q7.

Question	Scheme	Marks	AOs
(a)	Deduces that gradient of PA is $-\frac{1}{2}$	M1	2.2a
	Finding the equation of a line with gradient " $-\frac{1}{2}$ " and point $(7, 5)$ $y - 5 = -\frac{1}{2}(x - 7)$	M1	1.1b
	Completes proof $2y + x = 17$ *	A1*	1.1b
		(3)	
(b)	Solves $2y + x = 17$ and $y = 2x + 1$ simultaneously	M1	2.1
	$P = (3, 7)$	A1	1.1b
	Length $PA = \sqrt{(3-7)^2 + (7-5)^2} = (\sqrt{20})$	M1	1.1b
	Equation of C is $(x-7)^2 + (y-5)^2 = 20$	A1	1.1b
		(4)	
(c)	Attempts to find where $y = 2x + k$ meets C using $\overrightarrow{OA} + \overrightarrow{PA}$	M1	3.1a
	Substitutes their $(11, 3)$ in $y = 2x + k$ to find k	M1	2.1
	$k = -19$	A1	1.1b
		(3)	
(10 marks)			

(c)	Attempts to find where $y = 2x + k$ meets C via simultaneous equations proceeding to a 3TQ in x (or y) FYI $5x^2 + (4k - 34)x + k^2 - 10k + 54 = 0$	M1	3.1a
	Uses $b^2 - 4ac = 0$ oe and proceeds to $k = \dots$	M1	2.1
	$k = -19$	A1	1.1b
		(3)	

Notes:

(a)

M1: Uses the idea of perpendicular gradients to deduce that gradient of PA is $-\frac{1}{2}$. Condone $-\frac{1}{2}x$ if followed by correct work. You may well see the perpendicular line set up as $y = -\frac{1}{2}x + c$ which scored this mark

M1: Award for the method of finding the equation of a line with a changed gradient and the point $(7, 5)$

So sight of $y - 5 = \frac{1}{2}(x - 7)$ would score this mark

If the form $y = mx + c$ is used expect the candidates to proceed as far as $c = \dots$ to score this mark.

A1*: Completes proof with no errors or omissions $2y + x = 17$

(b)

M1: Awarded for an attempt at the key step of finding the coordinates of point P , ie for an attempt at solving $2y + x = 17$ and $y = 2x + 1$ simultaneously. Allow any methods (including use of a calculator) but it must be a valid attempt to find both coordinates. Do not allow where they start $17 - x = 2x + 1$ as they have set $2y = y$ but condone bracketing errors, eg $2 \times 2x + 1 + x = 17$

A1: $P = (3, 7)$

M1: Uses Pythagoras' Theorem to find the radius or radius ² using their $P = (3, 7)$ and $(7, 5)$. There must be an attempt to find the difference between the coordinates in the use of Pythagoras

A1: $(x-7)^2 + (y-5)^2 = 20$. Do not accept $(x-7)^2 + (y-5)^2 = (\sqrt{20})^2$

(c)

M1: Attempts to find where $y = 2x + k$ meets C .

Awarded for using $\overrightarrow{OA} + \overrightarrow{PA}$. $(11, 3)$ or one correct coordinate of $(11, 3)$ is evidence of this award.

M1: For a full method leading to k . Scored for either substituting their $(11, 3)$ in $y = 2x + k$

or, **in the alternative**, for solving their $(4k-34)^2 - 4 \times 5 \times (k^2 - 10k + 54) = 0 \Rightarrow k = \dots$ Allow use of a calculator here to find roots. Award if you see use of correct formula but it would be implied by $\pm$ correct roots

A1: $k = -19$ only

Alternative I

M1: For solving $y = 2x + k$ with their $(x-7)^2 + (y-5)^2 = 20$ and creating a quadratic eqn of the form $ax^2 + bx + c = 0$ where both b and c are dependent upon k . The terms in x^2 and x must be collected together or implied to have been collected by their correct use in " $b^2 - 4ac$ "

FYT the correct quadratic is $5x^2 + (4k-34)x + k^2 - 10k + 54 = 0$

M1: For using the discriminant condition $b^2 - 4ac = 0$ to find k . It is not dependent upon the previous M and may be awarded from only one term in k .

$(4k-34)^2 - 4 \times 5 \times (k^2 - 10k + 54) = 0 \Rightarrow k = \dots$ Allow use of a calculator here to find roots.

Award if you see use of correct formula but it would be implied by $\pm$ correct roots

A1: $k = -19$ only

Alternative II

M1: For solving $2y + x = 17$ with their $(x-7)^2 + (y-5)^2 = 20$, creating a 3TQ and solving.

M1: For substituting their $(11, 3)$ into $y = 2x + k$ and finding k

A1: $k = -19$ only

.....
Other method are possible using trigonometry.